HyBench3D: A Leakage-free Benchmark of 3D CNN, Transformer, and Hybrid Networks for Alzheimer's Disease Classification in 3D MRI

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***Abstract*—In this study, we propose HyBench3D, a leakage-free benchmark that evaluates whether hybrid networks combining a three-dimensional convolutional neural network (3D CNN) with a vision transformer (ViT) improve Alzheimer's disease (AD) classification in 3D MRI. The 3D CNN extracts local features of AD-related abnormalities with its inductive bias, while the transformer models long-range relationships among distant brain regions. Many previous studies report accuracies above 95%, but most of them split data at the level of 2D slices, so that images of the same patient appear in both training and test sets. In HyBench3D, all networks share the same preprocessing, classifier head, and subject-level five-fold splits, and their differences are tested statistically. In this experiment, we used 1,132 3D T1-weighted MRI images of 395 subjects from the Alzheimer's Disease Neuroimaging Initiative (ADNI) and compared six networks: 3D DenseNet-121, MedicalNet-pretrained 3D ResNet-18, 3D ViT, Swin Transformer, and two hybrids (CNN–ViT and Swin–ViT). For cognitively normal (CN) versus AD classification, the CNN–ViT hybrid achieved the highest balanced accuracy of 0.806, but it was statistically indistinguishable from 3D DenseNet-121 (0.800, $p = 0.54$) while using twice the parameters. Removing the CNN branch decreased the macro-F1 score in all five folds, and an age-and-sex-only baseline remained at chance level. For CN versus mild cognitive impairment (MCI) versus AD classification, the best balanced accuracy was 0.549 against a chance level of 0.333. This study demonstrates that, under leakage-free evaluation, the convolutional branch carries the discriminative signal, and the transformer branch adds no measurable value at the scale of a typical ADNI cohort.**

***Index Terms*—Alzheimer's disease, 3D MRI, 3D CNN, vision transformer, hybrid network, data leakage, explainable AI**

I. Introduction

Convolutional neural networks (CNNs) have established a dominant performance in computer vision [1], and they have been widely applied to medical image analysis [2]. For 3D medical images, a 3D CNN can learn native volumetric representations, and densely connected architectures such as DenseNet [3] train efficiently on small cohorts. However, few large 3D databases are available for pretraining [4], and 3D networks require large memory and computation. Especially, the strong inductive bias of convolution restricts each layer to a limited receptive field.

Meanwhile, transformer networks [5] have been widely used in computer vision. The vision transformer (ViT) [6] models the relationships among all image patches with self-attention, so its receptive field covers the whole image. ViT is competitive with CNNs on very large databases, but it tends to underperform on small datasets because it lacks the inductive bias of convolution. The Swin Transformer [7], [8] partly restores locality with shifted windows.

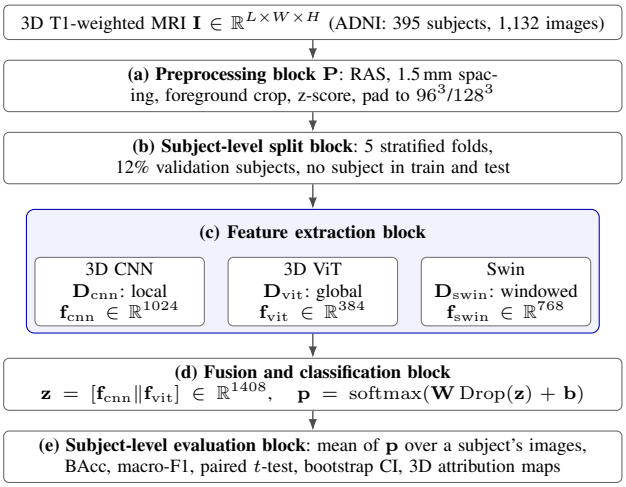


Fig. 1. The overall framework of HyBench3D. Every network shares blocks (a), (b), (d), and (e); only the backbone in block (c) changes. The hybrid networks concatenate the features of two backbones.

To take advantage of both, hybrid networks combining CNN and transformer have been proposed for Alzheimer's disease (AD) classification [9]–[11], and many of them report accuracies above 95%. However, two problems limit the conclusions of these studies. First, many of them split the data at the level of 2D slices, so that slices of the same patient appear in both training and test sets. Yagis *et al.* [12] showed that, on the same data and network, this data leakage alone raises the accuracy by about 28 points, and more than half of the reviewed CNN studies may suffer from it [13], [14]. Even 3D networks can leak through longitudinal images of the same subject [15]. Second, hybrid networks are rarely compared with equally tuned single networks on identical splits with statistical tests, so the contribution of the fusion itself remains unknown.

AD is a progressive neurodegenerative disease that causes memory loss and gradually impairs daily functioning [16]. Its progression involves structural changes such as hippocampal shrinkage, ventricular enlargement, and cortical atrophy [17], which are visible in 3D T1-weighted MRI (Fig. 2). The hippocampal and ventricular changes occur in local regions, which suits a CNN with inductive bias. On the other hand, the cortical atrophy is distributed throughout the brain, which

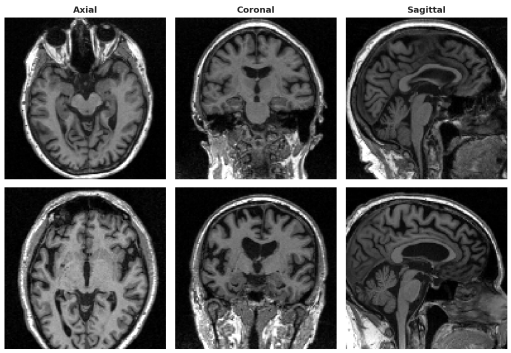


Fig. 2. Comparison of a cognitively normal (CN) brain (top) and the structural changes of an Alzheimer's disease (AD) brain (bottom) in axial, coronal, and sagittal planes. The AD brain shows enlarged ventricles and atrophy of the medial temporal lobe.

suits a transformer with a wide receptive field. Accordingly, a CNN–transformer hybrid is a natural candidate for AD classification, and the open question is whether it measurably improves on a strong CNN under honest evaluation.

In this study, we propose HyBench3D, a leakage-free benchmark of 3D CNN, transformer, and hybrid networks for AD classification in 3D MRI. Our goal is to measure whether a transformer branch improves a 3D CNN for cognitively normal (CN) versus AD classification and for CN versus mild cognitive impairment (MCI) versus AD classification. The overall framework of HyBench3D is shown in Fig. 1. The main contributions of this study are as follows.

First, HyBench3D evaluates six 3D networks with identical preprocessing, classifier head, imbalance handling, and subject-level splits, and reports majority-class and demographic baselines, fold-paired tests, bootstrap intervals, and network complexity.

Second, through a branch ablation, we show that adding a ViT branch to a 3D CNN doubles the number of parameters (from 11.24M to 23.55M) without a measurable gain ($p = 0.54$), whereas replacing the CNN branch decreases the macro-F1 score by 0.14 in every fold.

Third, we report two failure modes that are rarely documented: the instability of a medically pretrained 3D ResNet, and the collapse of a 3D ViT trained from scratch that never predicts the majority class.

Fourth, we visualize the networks with eight 3D attribution methods, show where they disagree, and provide a checklist for evaluating hybrid networks.

II. Related Works

A. Transformer and Hybrid Networks for Vision

SwinUNETR [8] extends the shifted-window transformer [7] to 3D medical images. Hybrid networks combine the locality of convolution with the global receptive field of self-attention; for AD, a CNN–Swin hybrid reported 95.34% on 2D OASIS slices [9], and a ViT–GRU network up to 99.69% [10].

B. 3D Medical Image Analysis and Data Leakage

3D CNNs learn volumetric representations directly from MRI volumes [18], and transfer learning from medical 3D databases is provided by MedicalNet [4]. However, the evaluation protocol strongly affects the reported performance. Studies that use slice-level splits or curated 2D images report a median accuracy of 99.0% [9], [10], [19]–[24], whereas studies with subject-level splits report 60–90% [13], [25]. Recent reviews identify data leakage as the most recurrent source of performance inflation in this field [14].

C. Alzheimer's Disease Classification

Under subject-level evaluation, Wen *et al.* [13] reported 85–88% for CN versus AD with 3D CNNs, and Saoud and AlMarzouqi [25] reported 90% for CN versus AD and 60% for the three-class task with an ensemble of 138 regional 3D ViTs. A concurrent study reports 72.3% on OASIS-1 with a multi-modal 3D CNN [26]. M3T [11] combines 3D and 2D CNNs with a transformer, was trained on 4,786 ADNI images, and was tested on independent ADNI, AIBL, and OASIS sets. It reached 93.2% on ADNI, 1.2 points above a 3D DenseNet-201, from a single run without significance tests. In our work, we focus on whether the transformer branch itself adds value over a matched 3D CNN.

III. Methods

A. Overview of HyBench3D

HyBench3D consists of five main blocks (Fig. 1): 1) a preprocessing block to standardize the 3D MRI images, 2) a subject-level split block to prevent data leakage, 3) a feature extraction block with interchangeable 3D CNN and transformer backbones, 4) a fusion and classification block with a shared head, and 5) a subject-level evaluation block. Only the backbone in block 3) changes between networks.

B. Preprocessing Block

Given a 3D T1-weighted MRI image $\mathbf{I}$, the preprocessing block $\mathbf{P}$ computes the standardized input $\mathbf{X}$:

$$\mathbf{X} = \mathbf{P}(\mathbf{I}) = \mathbf{R}\big(\mathbf{Z}(\mathbf{C}(\mathbf{S}(\mathbf{O}(\mathbf{I}))))\big) \in \mathbb{R}^{L\times W\times H}, \qquad (1)$$

where $\mathbf{O}$ reorients the image to the RAS axes, $\mathbf{S}$ resamples it to $1.5 \times 1.5 \times 1.5\,\text{mm}$, $\mathbf{C}$ crops the foreground, $\mathbf{Z}$ applies z-score normalization over non-zero voxels, and $\mathbf{R}$ pads or crops it to $L = W = H$. We use $L = 128$ for the DenseNet and ResNet networks and $L = 96$ for all transformer and hybrid networks. No skull stripping or registration is applied.

C. Subject-level Split Block

Let $\mathcal{D} = \{(\mathbf{I}_i, y_i, s_i)\}_{i=1}^{N}$ be the images, where y_i is the label and s_i the subject. The subjects are divided into five stratified folds (seed 42), and fold k is tested after training on the remaining folds, so that

$$\{s_i : i \in \mathcal{T}_k^{\text{train}}\} \cap \{s_i : i \in \mathcal{T}_k^{\text{test}}\} = \varnothing. \qquad (2)$$

Inside each training partition, 12% of the subjects are held out as a validation set for early stopping and checkpoint selection, so the test fold is never used for model selection.

D. Feature Extraction Block

Each backbone $\mathbf{D}$ maps $\mathbf{X}$ to a feature vector $\mathbf{f}$. The 3D CNN branch is a DenseNet-121 [3] trained from scratch with global average pooling:

$$\mathbf{f}_{\text{cnn}} = \mathbf{D}_{\text{cnn}}(\mathbf{X}) \in \mathbb{R}^{1024}. \tag{3}$$

The ViT branch [6] divides $\mathbf{X}$ into $N_p = LWH/p^3$ non-overlapping patches of size p^3 ($p = 16$, $N_p = 216$), encodes them with 6 transformer layers (6 heads, hidden size $d = 384$, MLP size 1536), and averages the output tokens $\mathbf{T} \in \mathbb{R}^{N_p \times d}$:

$$\mathbf{f}_{\text{vit}} = \frac{1}{N_p} \sum_{n=1}^{N_p} \mathbf{T}_n \in \mathbb{R}^{384}. \tag{4}$$

The Swin branch uses the SwinUNETR encoder [8] (embedding dimension 48, depths $(2, 2, 2, 2)$, heads $(3, 6, 12, 24)$, window 7^3) with global average pooling of the deepest stage, $\mathbf{f}_{\text{swin}} \in \mathbb{R}^{768}$. The ResNet-18 branch is initialized with MedicalNet weights [4], $\mathbf{f}_{\text{res}} \in \mathbb{R}^{512}$.

E. Fusion and Classification Block

A hybrid network applies two backbones to the same input and concatenates their features. For the CNN–ViT hybrid,

$$\mathbf{z} = [\mathbf{f}_{\text{cnn}} \,\|\, \mathbf{f}_{\text{vit}}] \in \mathbb{R}^{1408}, \tag{5}$$

and the Swin–ViT hybrid replaces $\mathbf{f}_{\text{cnn}}$ with $\mathbf{f}_{\text{swin}}$. A single network uses $\mathbf{z} = \mathbf{f}$. The shared head computes the class probabilities

$$\mathbf{p} = \text{softmax}\big(\mathbf{W}\, \text{Drop}_{0.2}(\mathbf{z}) + \mathbf{b}\big) \in \mathbb{R}^K, \tag{6}$$

where $K = 2$ or 3. For the binary task we use the class-weighted cross-entropy with label smoothing $\epsilon = 0.1$ [27],

$$\ell = -\sum_{c=1}^{K} w_c\, q_c \log p_c, \qquad q_c = (1 - \epsilon)\, y_c + \epsilon/K, \tag{7}$$

with $w_c = n/(K\, n_c)$, where n_c is the number of training images of class c. For the three-class task we use the focal loss [28] $\mathcal{L} = (1 - e^{-\ell})^\gamma\, \ell$ with $\gamma = 2$ and $\epsilon = 0$. In addition, every network draws training batches with a class-balanced sampler, $P(i) \propto 1/n_{y_i}$.

F. Subject-level Evaluation Block

Because a subject contributes 2.9 images on average, the probabilities of its images $I_s = \{i : s_i = s\}$ are averaged:

$$\hat{y}_s = \arg\max_c \frac{1}{|I_s|} \sum_{i \in I_s} p_c(\mathbf{I}_i). \tag{8}$$

We report the balanced accuracy $\text{BAcc} = \frac{1}{K} \sum_c \text{TP}_c/n_c$, the macro-F1 score, accuracy, sensitivity, and specificity (AD as the positive class). Accuracy and macro-F1 are given as mean $\pm$ standard deviation over the five folds, and BAcc with a 95% subject-level bootstrap confidence interval (10,000 resamples). Networks are compared with a paired t-test on the fold-wise macro-F1 scores.

IV. Experiments

A. Experimental Dataset

In this study, we acquired 3D T1-weighted MPRAGE images from ADNI [29], [30] (ADNI GO/2/3, 3T, pre-processed N3m NIfTI). After removing 18 off-protocol images, the dataset contains 1,132 images of 395 subjects: 206 CN (627 images), 80 MCI (182), and 109 AD (323). Each subject contributes up to six images acquired within two years (median one year), and each image inherits the ADNI research group of its subject. The groups are matched for age (CN 74.8 ± 7.1, MCI 73.9 ± 8.2, AD 74.7 ± 7.9 years; CN versus AD $p = 0.93$) and sex (53%, 46%, and 44% female; $p = 0.24$). The ADNI data were collected with the approval of the institutional review boards of the participating sites and are used by authorized investigators under the ADNI Data Use Agreement.

B. Implementation Details

We implemented HyBench3D with PyTorch [31] and MONAI [32]. All networks were trained with the AdamW optimizer [33] ($\beta_1 = 0.9$, $\beta_2 = 0.999$), linear warmup followed by cosine decay, mixed precision, a batch size of 2 with 4-step gradient accumulation (effective batch size of 8), and gradient clipping at 1.0 on one NVIDIA RTX 5060 Ti GPU (16 GB). The learning rate is 10^{-4}, except for the three-class Swin (5×10^{-5}) and the MedicalNet backbone, which is frozen for 5 epochs and then trained at 10^{-5}. The maximum number of epochs, warmup epochs, early-stopping patience, and weight decay are 80/5/15/0.01 for DenseNet-121, 60/5/12/0.01 for ResNet-18, 100/8/18/0.03 for the hybrids, 100/5/18/0.02 for Swin, and 100/8/18/0.05 for ViT. Training-time augmentation consists of left–right flips, random affine transforms ($\pm 10°$, $\pm 10\%$ scale, 8-voxel translation), Gaussian noise and smoothing, bias field, gamma (0.8–1.2), and intensity scaling and shifting.

C. Comparison Study Results

The quantitative performance for CN versus AD classification is presented in Table I. The CNN–ViT hybrid achieves the highest accuracy (0.822), macro-F1 (0.802), BAcc (0.806), and specificity (0.859). However, the 95% confidence interval of its BAcc almost coincides with that of 3D DenseNet-121 (0.752–0.846), which achieves a similar performance with less than half the parameters (11.24M versus 23.55M). The Swin–ViT hybrid and the pretrained ResNet-18 are about 12 points lower and much less stable, and the age-and-sex-only classifier remains at chance level, so the networks do not exploit demographic shortcuts.

For CN versus MCI versus AD classification (Table II), Swin and 3D DenseNet-121 are statistically tied ($p = 0.52$) at a BAcc of about 0.54–0.55, which is clearly above chance. The per-class F1 score is about 0.65 for CN and 0.55–0.61 for AD, but only 0.37–0.41 for MCI. As shown in Fig. 3(b)–(c), most MCI subjects are classified as CN or AD, which is consistent with the clinically intermediate nature of MCI. On the other hand, the 3D ViT trained from scratch never predicts CN (Fig. 3(d)), so its high MCI recall is an artifact

TABLE I

COMPARISON OF 3D CLASSIFICATION NETWORKS FOR CN VERSUS AD CLASSIFICATION (315 SUBJECTS). ACCURACY AND MACRO-F1: MEAN ± SD OVER FIVE SUBJECT-LEVEL FOLDS. BACC, SENSITIVITY (SENS.), AND SPECIFICITY (SPEC.): POOLED OUT-OF-FOLD PREDICTIONS WITH A 95% BOOTSTRAP CI. PARAMS AND GFLOPS: MEASURED FOR ONE INPUT VOLUME.

Model name	Input	Params	GFLOPs	Accuracy	Macro-F1	BAcc [95% CI]	Sens.	Spec.
3D DenseNet-121	128^3	11.24M	85.8	0.810±0.041	0.788±0.062	0.800 [0.752, 0.846]	**0.771**	0.830
3D ResNet-18 (MedicalNet)	128^3	32.99M	501.6	0.698±0.139	0.654±0.168	0.689 [0.635, 0.742]	0.670	0.709
Swin–ViT hybrid	96^3	20.37M†	55.5†	0.702±0.070	0.662±0.101	0.670 [0.615, 0.724]	0.569	0.772
CNN–ViT hybrid	96^3	23.55M	41.9	**0.822**±0.057	**0.802**±0.062	**0.806** [0.757, 0.852]	0.752	**0.859**
Age + sex only (logistic)	–	–	–	*0.498±0.092*	*0.486±0.092*	*0.503*‡	–	–
Majority class (always CN)	–	–	–	*0.654*	*0.395*	*0.500*	*0.000*	*1.000*

†Released concatenation variant; the reported run used an earlier ViT branch with extra cross-attention. ‡Mean over folds.

TABLE II

COMPARISON OF NETWORKS FOR CN VERSUS MCI VERSUS AD CLASSIFICATION (395 SUBJECTS). CHANCE-LEVEL BACC IS 0.333.

Model name	Params	Macro-F1	BAcc [95% CI]	MCI recall
3D DenseNet-121	11.25M	0.524±0.071	0.537 [0.485, 0.588]	**0.400**
Swin	8.06M	**0.537**±0.078	**0.549** [0.499, 0.601]	0.375
3D ViT (scratch)	12.30M	0.256±0.012	0.409 [0.363, 0.454]	0.713
Age + sex only	–	*0.371±0.018*	*0.380*	–
Majority class	–	*0.229*	*0.333*	*0.000*

TABLE III

ABLATION OF THE CNN–VIT HYBRID FOR CN VERSUS AD CLASSIFICATION. Δ: FOLD-PAIRED DIFFERENCE IN MACRO-F1 FROM THE FULL HYBRID.

Model	Params	Macro-F1	Δ	p	Folds lost
w/o ViT branch	11.24M	0.788	−0.014	0.54	3/5
CNN branch → Swin	20.37M	0.662	−0.140	0.11	5/5
CNN–ViT (full)	23.55M	**0.802**	–	–	–

rather than evidence of skill. The pure transformer cannot learn meaningful representations from a few hundred subjects [6], and the double imbalance correction pushes it toward the minority classes.

D. Ablation Study Results

To evaluate the degree to which each branch of the CNN–ViT hybrid affects the performance, we compared it with two networks: 1) the hybrid without the ViT branch, which is 3D DenseNet-121, and 2) the hybrid with the CNN branch replaced by Swin, which is the Swin–ViT hybrid. Table III shows the results.

Removing the ViT branch reduces the macro-F1 score by only 0.014. The paired t-test gives $t = 0.67$ with 4 degrees of freedom ($p = 0.54$), and the full hybrid classifies 259 of the 315 subjects correctly, against 255 for the CNN alone. On the other hand, replacing the CNN branch with Swin reduces the macro-F1 score by 0.140 and loses in all five folds (Fig. 3(a)). The t-test is not significant because of one extreme fold ($p = 0.11$), but the BAcc confidence intervals do not overlap (0.615–0.724 versus 0.757–0.852). The convolutional branch therefore carries most of the signal, although input size and schedule also differ, so this is not a strict single-factor ablation. In addition, the MedicalNet-pretrained ResNet-18 is bimodal across folds (three folds reach 0.78–0.82 macro-F1 and two fall to 0.449). Its frozen-backbone start, $10\times$ lower backbone learning rate, and short patience of 12 epochs make a premature early stop likely, although we have not yet confirmed this with a minimum-epoch guard.

E. Visualization Results

We visualize the decisions of the networks with Grad-CAM [34], Grad-CAM++ [35], Score-CAM, Integrated Gradients [36], SmoothGrad [37], occlusion, LIME [38], and attention rollout [39]. Fig. 4 shows that, on the same subject, the methods highlight different regions. For the Swin and Swin–ViT networks, the Integrated Gradients averaged over 16 AD subjects is nearly uniform across thirds of the volume (0.036–0.055), with mass on the cerebellum, orbits, and scalp rather than a medial-temporal focus. Similar shortcut cues have been reported for 3D MRI networks [15]. Therefore, without skull stripping and atlas registration, the attribution maps cannot confirm that the networks rely on AD-specific anatomy.

V. DISCUSSION

Our results lie in the same regime as other subject-level studies: 82% for CN versus AD against 85–90% [13], [25], and 58% for three classes against 60% [25]; the gap is plausibly due to our minimal preprocessing and 96^3 input. The baseline table of M3T [11] points in the same direction as our ablation: with four times our training data, adding a transformer to DenseNet-201 *lowered* the ADNI accuracy from 92.0% to 90.4%, and a pure 3D ViT was again the weakest network.

We therefore recommend the following checklist for hybrid networks: 1) split at the subject level; 2) report majority-class, chance, and demographic baselines; 3) compare a hybrid with each of its branches on identical folds; 4) use fold-paired tests and confidence intervals; 5) report fold-wise results; and 6) skull-strip inputs and use several attribution methods before anatomical claims.

This study has the following limitations. Standalone ViT and Swin were not trained for the binary task, and the CNN–ViT hybrid was not trained for the three-class task. The results come from a single seed and five folds, so a small benefit of hybridization cannot be ruled out. The labels are the ADNI

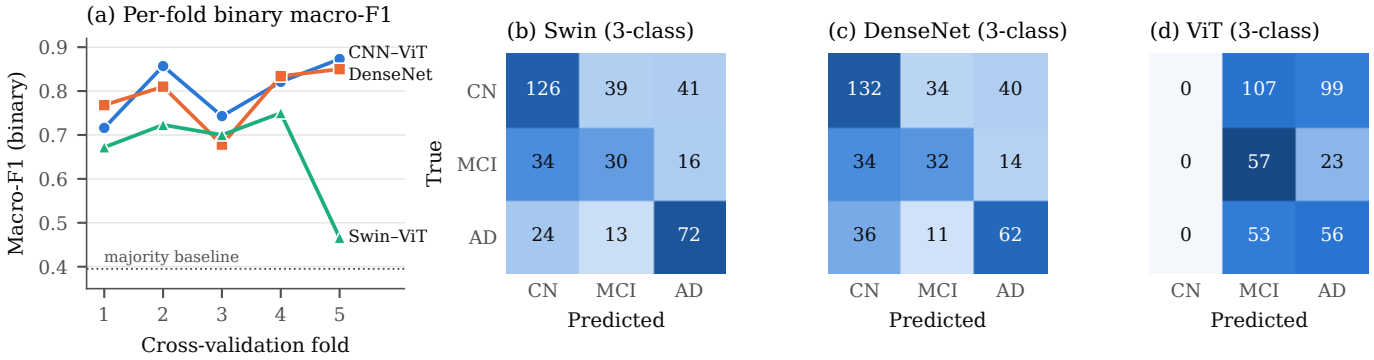


Fig. 3. (a) Fold-wise macro-F1 for CN versus AD classification on identical subject-level folds: the CNN–ViT hybrid and 3D DenseNet-121 alternate, whereas the Swin–ViT hybrid without a CNN branch is lower in every fold. (b)–(d) Pooled three-class confusion matrices (subject counts; shading is row-normalized).

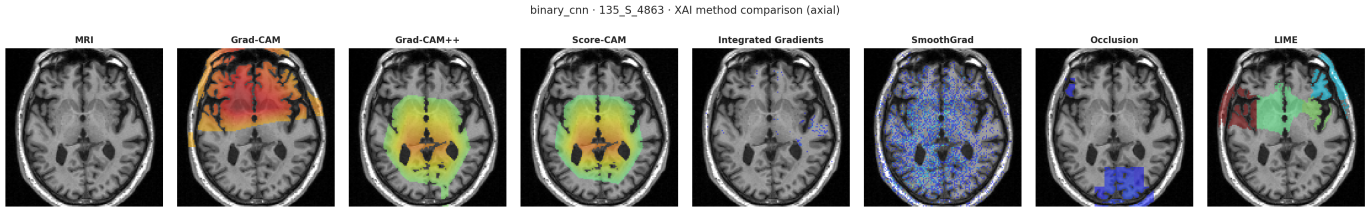


Fig. 4. Visualization results of eight 3D attribution methods for 3D DenseNet-121 on one AD subject (axial slice). Grad-CAM focuses on the frontal region, Grad-CAM++ and Score-CAM on the central region, occlusion on the occipital region and posterior skull, and LIME partly on the scalp.

enrolment groups, applied to images acquired within two years. Finally, the networks were evaluated only on ADNI.

VI. Conclusion

In this paper, we proposed HyBench3D, a leakage-free benchmark of 3D CNN, transformer, and hybrid networks for AD classification in 3D MRI. Experimental results show that the CNN–ViT hybrid achieves the highest scores for CN versus AD classification, but it is statistically indistinguishable from a 3D CNN with half the parameters, whereas removing the CNN branch consistently decreases the performance. The three-class task is limited by MCI, and the pure 3D ViT fails without pretraining. The visualization results show that attribution maps on non-skull-stripped images are not anatomically conclusive. In future work, we will complete the network grid with repeated cross-validation, add skull stripping and atlas-based attribution, and validate the networks on independent cohorts such as AIBL and OASIS.

Code Availability

The code, configurations, and subject-level split files will be made publicly available upon acceptance. The ADNI images must be obtained from ADNI under its Data Use Agreement.

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The authors used Anthropic Claude Sonnet 4.6 solely to check the grammar of the manuscript. All output was reviewed and approved by the authors, who take full responsibility for the content.

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REFERENCES

[1] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.

[2] G. Litjens, T. Kooi, B. E. Bejnordi, A. A. A. Setio, F. Ciompi, M. Ghafoorian, J. A. Van Der Laak, B. Van Ginneken, and C. I. Sánchez, "A survey on deep learning in medical image analysis," *Medical Image Analysis*, vol. 42, pp. 60–88, 2017.

[3] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 4700–4708.

[4] S. Chen, K. Ma, and Y. Zheng, "Med3D: Transfer learning for 3D medical image analysis," 2019.

[5] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 5998–6008.

[6] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations (ICLR)*, 2021.

[7] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 10 012–10 022.

[8] A. Hatamizadeh, V. Nath, Y. Tang, D. Yang, H. R. Roth, and D. Xu, "Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images," in *International MICCAI Brainlesion Workshop*. Springer, 2021, pp. 272–284.

[9] K. Velu and N. Jaisankar, "Design of a cnn–swin transformer model for alzheimer's disease prediction using mri images," *IEEE Access*, 2025.

[10] S. Mahim, M. S. Ali, M. O. Hasan, A. A. N. Nafi, A. Sadat, S. Al Hasan, B. Shareef, M. M. Ahsan, M. K. Islam, M. S. Miah *et al.*, "Unlocking the potential of xai for improved alzheimer's disease detection and classification using a vit-gru model," *IEEE Access*, vol. 12, pp. 8390–8412, 2024.

[11] J. Jang and D. Hwang, "M3T: Three-dimensional medical image classifier using multi-plane and multi-slice transformer," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 20 718–20 729.

[12] E. Yagis, S. W. Atnafu, A. García Seco de Herrera, C. Marzi, R. Scheda, M. Giannelli, C. Tessa, L. Citi, and S. Diciotti, "Effect of data leakage in brain MRI classification using 2D convolutional neural networks," *Scientific Reports*, vol. 11, no. 1, p. 22544, 2021.

[13] J. Wen, E. Thibeau-Sutre, M. Diaz-Melo, J. Samper-González, A. Routier, S. Bottani, D. Dormont, S. Durrleman, N. Burgos, and O. Colliot, "Convolutional neural networks for classification of alzheimer's disease: Overview and reproducible evaluation," *Medical Image Analysis*, vol. 63, p. 101694, 2020.

[14] V. M. Young, S. Gates, L. Y. Garcia, and A. Salardini, "Data leakage in deep learning for Alzheimer's disease diagnosis: A scoping review of methodological rigor and performance inflation," *Diagnostics*, vol. 15, no. 18, p. 2348, 2025.

[15] D. J. Rumala, "How you split matters: Data leakage and subject characteristics studies in longitudinal brain MRI analysis," in *Clinical Image-Based Procedures, Fairness of AI in Medical Imaging, and Ethical and Philosophical Issues in Medical Imaging (MICCAI 2023 Workshops)*, ser. Lecture Notes in Computer Science, vol. 14242. Springer, 2023.

[16] P. Scheltens, K. Blennow, M. M. Breteler, B. de Strooper, G. B. Frisoni, S. Salloway, and W. M. Van der Flier, "Alzheimer's disease," *The Lancet*, vol. 388, no. 10043, pp. 505–517, 2016.

[17] G. B. Frisoni, N. C. Fox, C. R. Jack Jr, P. Scheltens, and P. M. Thompson, "The clinical use of structural MRI in alzheimer disease," *Nature Reviews Neurology*, vol. 6, no. 2, pp. 67–77, 2010.

[18] K. Bäckström, M. Nazari, I. Y.-H. Gu, and A. S. Jakola, "An efficient 3D deep convolutional network for alzheimer's disease diagnosis using MR images," in *IEEE 15th International Symposium on Biomedical Imaging (ISBI)*. IEEE, 2018, pp. 149–153.

[19] M. H. Alshayeji, "Alzheimer's disease detection and stage identification from magnetic resonance brain images using vision transformer," *Machine Learning: Science and Technology*, vol. 5, no. 3, p. 035011, 2024.

[20] A. El-Assy, H. M. Amer, H. Ibrahim, and M. Mohamed, "A novel cnn architecture for accurate early detection and classification of alzheimer's disease using mri data," *Scientific Reports*, vol. 14, no. 1, p. 3463, 2024.

[21] D. AlSaeed and S. F. Omar, "Brain mri analysis for alzheimer's disease diagnosis using cnn-based feature extraction and machine learning," *Sensors*, vol. 22, no. 8, p. 2911, 2022.

[22] A. Mora-Rubio, M. A. Bravo-Ortíz, S. Q. Arredondo, J. M. S. Torres, G. A. Ruz, and R. Tabares-Soto, "Classification of alzheimer's disease stages from magnetic resonance images using deep learning," *PeerJ Computer Science*, vol. 9, p. e1490, 2023.

[23] D. Kurniasari, D. P. Muhammad, A. J. Junaidi, and A. Faisol, "Transforming alzheimer's disease diagnosis: Implementing vision transformer (vit) for mri images classification," *Journal of Information and Communication Technology (JICT)*, vol. 24, no. 1, pp. 130–152, 2024.

[24] A. Pandey, A. Pandey, and S. Poudel, "Leveraging vision transformers for early detection of alzheimer's disease through mri image analysis," *Kathford Journal of Engineering and Management*, vol. 4, no. 1, pp. 31–47, 2025.

[25] L. S. Saoud and H. AlMarzouqi, "Explainable early detection of alzheimer's disease using rois and an ensemble of 138 3d vision transformers," *Scientific Reports*, vol. 14, no. 1, p. 27756, 2024.

[26] M. Sifat, S. Akter, A. Islam, M. E. Hamid, A. S. M. Miah, N. Hassan, M. A. Rahim, and J. Shin, "3D MRI-based Alzheimer's disease classification using multi-modal 3D CNN with leakage-aware subject-level evaluation," arXiv:2603.17304, 2026.

[27] R. Müller, S. Kornblith, and G. E. Hinton, "When does label smoothing help?" in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.

[28] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2980–2988.

[29] C. R. Jack Jr, M. A. Bernstein, N. C. Fox, P. Thompson, G. Alexander, D. Harvey, B. Borowski, P. J. Britson, J. L Whitwell, C. Ward *et al.*, "The alzheimer's disease neuroimaging initiative (ADNI): MRI methods," *Journal of Magnetic Resonance Imaging*, vol. 27, no. 4, pp. 685–691, 2008.

[30] R. C. Petersen, P. S. Aisen, L. A. Beckett, M. C. Donohue, A. C. Gamst, D. J. Harvey, C. R. Jack Jr, W. J. Jagust, L. M. Shaw, A. W. Toga, J. Q. Trojanowski, and M. W. Weiner, "Alzheimer's disease neuroimaging initiative (ADNI): clinical characterization," *Neurology*, vol. 74, no. 3, pp. 201–209, 2010.

[31] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 8026–8037.

[32] M. J. Cardoso, W. Li, R. Brown, N. Ma, E. Kerfoot, Y. Wang, B. Murrey *et al.*, "MONAI: An open-source framework for deep learning in healthcare," 2022.

[33] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.

[34] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626.

[35] A. Chattopadhay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks," in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2018, pp. 839–847.

[36] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *International Conference on Machine Learning (ICML)*, 2017, pp. 3319–3328.

[37] D. Smilkov, N. Thorat, B. Kim, F. Viégas, and M. Wattenberg, "SmoothGrad: removing noise by adding noise," 2017.

[38] M. T. Ribeiro, S. Singh, and C. Guestrin, ""why should I trust you?": Explaining the predictions of any classifier," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 1135–1144.

[39] S. Abnar and W. Zuidema, "Quantifying attention flow in transformers," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 4190–4197.